

AHU-01 SEQUENCE OF OPERATION (FINAL)

Modes: (highest priority to lowest)

The unit shall control to one of five modes:

1. Maintenance Off
2. Safety Mode
3. Soft Start
4. Occupied Mode
5. Bypass Mode (DESIGN NOTE: Per RFI #23 - Not in Scope)
6. Unoccupied Mode

Maintenance Off Mode:

The Air Handler shall stop all controller functions and close all dampers and valves. All sensors will continue to report current condition. This is a manually selected mode and cannot be scheduled. NOTE: All required applicable safety procedures MUST still be used. Hardwired safeties will still actuate unless locked out via those safety procedures.

Safety alarms will be generated but may not cause functions if physically locked out. No other alarms are initiated during a maintenance off. An alert will be generated when the unit is placed in Maintenance Off and when returned to normal operation.

Safety Mode:

The Air Handler is equipped with safeties for Static Pressure on Supply and Return Ducts, a Fire Alarm Interface (which can be interfaced locally to Smoke Detectors if the Fire Alarm system can support this); and Low Temperature Detection (Freeze/Defrost).

If a safety device indicates a condition the controller shall enter the Safety Mode, while the hardwired functions will cause:

- The SA Fans and RA Fan will stop
- The OA, MCA and EA dampers shall close
- The RA damper shall open

If the type of safety device is a Low Temperature condition then additionally will perform these actions:

- The chilled water valve shall modulate fully open
- The pre-heat valve shall fully open (normally open)
- Circulation pump will run continuously
- In all Safety conditions an alarm will be generated with the initiating device clearly indicated. All safety devices are electrically hardwired and do not rely on the controller for operation, to stop operation completely, or at stop last position. The controller monitors safety devices only for indication. The chill water valve is the only exception, and it will be commanded open if required.

Soft Start:

When the unit is started and placed in operation, transitioning from Maintenance Off, Safety Mode, or power restoration, before entering Occupied or Unoccupied mode then following shall be confirmed and completed:

- The minimum outside air (OA) damper shall open to its minimum setpoint over a 5 minute (adj.) period.
- The economizer shall be disabled for five minutes (adj.) before being released to automatic control.
- SA and RA Fans shall start at minimum speed and shall hold there for 60 seconds. After that period they shall be released to ramp per programming.

Once all delays are satisfied then the unit shall be released to the scheduled mode.

Occupied Mode:

The unit shall be in Occupied Mode when:

- Scheduled from the Front End, based on existing schedules.
- OR unit mode control is overridden.

When the unit is in occupied mode the EA damper shall be controlled per the Exhaust Air sequence described below, the RA damper shall be open, and OA damper shall be closed and the Min OA damper shall modulate to maintain the Min OA flow setpoint.

Supply Air (SA) Fan:

The SA fans shall modulate speed in unison to maintain static pressure setpoint of 1.0" WC (adj.) at DPT-1 over a period of five minutes (adj.). To prevent short cycling, the SA fans shall have a minimum runtime of 5 minutes (adj.) and a minimum off time of 30 seconds. The SA fan is two parallel fans, but is commanded and controlled as a single unit, but are capable of running independently in a failure situation. If a fault or failure of one of the fans occurs, that fan shall be locked out until reset, the remaining fan shall ramp to maintain the S.P. setpoint.

Alarms shall be provided as follows:

- SA Fan Failure or locally switched Off: Commanded on, but the status is off. This will be noted as Fan A or B.
- SA Fan In Hand: Commanded off, but the status is on. This will be noted as Fan A or B.
- SA Duct Pressure Above or Below Setpoint by 10% (adj.).

Local-remote settings shall be provided as part of the variable frequency drive through the drive's keypad for supply fan. In the local mode, the supply/return fan's speed shall be controlled through a manual speed control located at the drive control panel. In the remote mode, the supply/return fan's speed shall be controlled by the BAS.

Volume Calculation:

Air volume shall be calculated at two points using the Air Flow Velocity sensors FS-1, FS-2, and the temperature sensors TS-1 and TS-2. Air volume shall be calculated by the formula $FS \times A$ in $PPH \times A$ in ft^2 (at sensor location) $\times$ temperature compensation as noted from the chart below. Note: flow sensors are monitor only points and do not control fans.

Air Temperature (°F)	Volume Correction Factor
32	0.93
40	0.94
50	0.95
60	0.96
70	0.97
80	0.98
90	0.99
100	1.00
110	1.01
120	1.02

Return Air (RA) Fan:

The RA fan shall start and run whenever the supply fan is indexed to run, in sequence to SA fan. RA fan shall be modulated to maintain RA setpoint at DPT-2 to 0.15" W.C. (adj.). (The pickup point for this

sensor DPT-2 shall be verified during Test & Balance. The RA fan shall have a ramp time of 5 min (adj.). To prevent short cycling, the RA fan shall have a minimum runtime of 5 minutes (adj.) and a minimum off time of 30 seconds.

The return fan speed shall not drop below 20% (adj.) to provide adequate motor cooling and prevent fan stalling.

Alarms shall be provided as follows:

- RA Fan Failure or locally switched Off: Commanded on, but the status is off.
- RA Fan In Hand: Commanded off, but the status is on.
- RA Static Pressure: If the RA static pressure is greater than setpoint of 0.2" W.C. (adj.) by 10% (adj.).

Heat Circulation Pump:

The Pump will run if the OA temperature is below 45°F (adj.). To prevent short cycling, the pump shall have a minimum runtime of 5 minutes (adj.).

Alarms shall be provided as follows:

- Pump Failure or locally switched Off: Commanded on, but the status is off.

Pre-Heat Hot Water Valve Control:

The controller shall measure the Mixed Air temperature and modulate the Heat Valve to maintain a Supply Air temperature setpoint of 55°F (adj.).

The heat valve loop shall be enabled whenever:

- OA temperature is below 35°F regardless of AHU status
- OR The AHU SA fan status is on
- AND The OA temperature is below 52°F (adj.).

MA Temperature:

The controller shall monitor the MA temperature. Alarms shall be provided as follows:

- High MA Temp: If the MA temperature is greater than 80°F (adj.).
- Low MA Temp: If the MA temperature is less than 30°F (adj.).

Cooling Coil Control:

The controller shall measure the SA temperature and modulate the cooling coil valve to maintain its cooling setpoint of 55°F (adj.).

The cooling valve shall be enabled whenever:

- OA temperature is greater than 80°F (adj.).
- AND The economizer is disabled or fully open.
- AND The SA fan status is on.
- AND The heating is not active.

If the economizer is modulating, but not maintaining the Mixed Air setpoint after 5 minutes (adj.) then the cooling coil will be modulated to maintain the cooling setpoint.

Alarms shall be provided as follows:

- High SA Temp: If the SA temperature is 5°F (adj.) greater than setpoint.
- Abnormal Operation: If the Economizer and Cooling at the same time for more than 30 minutes (adj.).

SA Temperature:

The controller shall monitor the SA temperature. Alarms shall be provided as follows:

- High SA Temp: If the SA temperature is greater than 120°F (adj.).
- Low SA Temp: If the SA temperature is less than 45°F (adj.).

Exhaust Air (EA) Damper Control: Relief aid

The controller shall measure building static pressure at DPT-4, (a dual action sensor, with range of -0.4" to 0.4" WC) to control the Exhaust Air Damper. The EA Damper will remain closed until the pressure is above 0.0" (adj.) WC, and then shall modulate as needed to maintain a positive building pressure 0.0" WC.

There will not be an active reset schedule.

Alarms shall be provided as follows:

- High Building Static Pressure: If the static pressure is above .30" WC (adj.).
- Low Building Static Pressure: If the static pressure is below 0.01" WC (adj.).

Economizer:

The controller shall measure the MA temperature at TS-3 and modulate the OA, exhaust, and return dampers in sequence to maintain a setpoint of 2°F (adj.) less than the SA temperature setpoint. The RA damper will inversely modulate according to the OA damper.

The economizer shall be enabled whenever:

- The MinOA is fully open.
- OA temperature is less than 55°F (adj.) AND OA temperature is greater than 57°F (adj.).
- AND The SA fan status is on
- AND The preheat coil has been off continuously for at least 10 minutes (adj.)

- The economizer shall be disabled whenever:
- OR the Low Temperature (Freeze/Defrost) is on
- OR loss of supply fan on status.
- OR if unit is commanded off.
- OR OA temperature is greater than 77°F (adj.) OR OA temperature is less than 57°F (adj.) (DESIGN NOTE: Much needs to be done to determine true low and OA velocity greater than 20 ft/min. (adj.). The Economizer (OA) damper and exhaust air dampers shall close and the return air damper shall open.

Humidification Note: Humidifier will be disabled in the program until activated by DPW

The controller shall measure the RA humidity to modulate the humidifier control valve to maintain return air humidity at 30 % (adj.) as indicated by HS-1.

Humidification shall be enabled whenever:

- The SA fan status is on.
- AND The OA temperature is below 45°F (adj.).
- AND The chilled water valve is closed.
- AND steam is available from the local steam generator (Future project).

Graphics will show steam availability feeding to the Humidifier.

SA and RA Humidity:

The controller shall monitor the humidity. Supply Air is a monitored reference point only. Alarms shall be provided as follows:

- Low Humidity: If the RA humidity is less than 25% RH (adj.).
- High Humidity: If the RA humidity is greater than 75% RH (adj.).

Minimum OA Control:

MOA damper will be modulated via Flow station to maintain MinOA Flow Setpoint (9540 CFM (adj.), occupied) at all times, except during a safety condition or in economizer operation. (The MinOA Damper is fully open in economizer operation.)

If the unit is not in economizer mode, and the OA setpoint is not achieved (Low OA flow) and the MinOA damper remains fully open for more than 5 min, then the Return air damper shall slowly modulate to the closed position to increase OA and allow the MinOA damper to regain control of OA volume. If the MinOA damper command is less than 10% and the OA setpoint is maintained for greater than 5 minutes the return damper shall slowly modulate toward the open position.

(DESIGN NOTE: These low conditions are not anticipated but addressed if the balance contractor determines an OA setpoint significantly lower than design.)

RA Temperature:

The controller shall monitor the RA. Alarms shall be provided as follows:

- High RA Temp: If the RA temperature is greater than 80°F (adj.).
- Low RA Temp: If the RA temperature is less than 45°F (adj.).

Bypass Mode:

If the outside air is compromised and full mode is required then an operator override to Bypass Mode will disable the economizer, the OA, MinOA, and EA dampers will fully close and the RA damper will fully open.

(DESIGN NOTE: Bypass has been determined to be a future requirement and not in scope for this project. A HVAC Shutdown button is installed but not terminated or programmed in this project. When implemented this should be a Safety condition and should move up in priority and will override occupied and all OA requirements. The RA Fan speed signal will be equal to the SA Fan speed signal plus or minus an offset (adj.), and scale factor (adj.) as set by the balance contractor. These should be a speed percentage and not a calculated value as this is a Safety condition.)

Unoccupied Mode:

The unit shall be in Unoccupied Mode when:

- Scheduled from the Front End.
- OR unit mode control is overridden.

Shall follow the same functional operation as described in the Occupied Mode. Setpoints shall be set as noted:

Heating Setpoint: 50°F (adj.)

Cooling Setpoint: 50°F (adj.)

(NOTE: The unit does not go Unoccupied currently, setpoints are the same as Occupied. Provide setpoints for future use.)

SA Duct Static: 1"WC (adj.)

MinOA Flow Setpoint: 4820 CFM (adj.)

External Setpoints

The AHU will accept external setpoints as shown below to optimize Supply Air Static Pressure and Supply Air Temperature setpoint as calculated from remote terminal devices. These points shall include:

- 01_Fan_Run_Input - This output will be linked to the supply fan status point in order for the setpoint calculation to occur.
- 01_SA_Setpoint_Req - This input is used for the requested SA Temp Setpoint
- 01_SA_Static_SP_Req - This input is used for the requested SA static setpoint

END OF SEQUENCE



NO.	REVISION	DATE	DESCRIPTION
1	ISSUED FOR PERMIT	08/01/2010	ISSUED FOR PERMIT
2	REVISION 1	08/01/2010	REVISION 1
3	REVISION 2	08/01/2010	REVISION 2
4	REVISION 3	08/01/2010	REVISION 3
5	REVISION 4	08/01/2010	REVISION 4
6	REVISION 5	08/01/2010	REVISION 5
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8	REVISION 7	08/01/2010	REVISION 7
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